Frontier



R&D application
2026

General Application

The General Application applies to everyone; all applicants should complete this section

Public section

The content in this section (answers to questions 1(a) - (e)) will be made public on the Frontier GitHub repository after we announce funding decisions in the summer and fall. Include as much detail as possible but omit sensitive and proprietary information.

Organization name

University of Cambridge and Woods Hole Oceanographic Institution

Organization location (we welcome applicants from anywhere in the world)

Cambridge, UK and Woods Hole, MA

Name(s) of primary point(s) of contact for this application

John Taylor (PI, University of Cambridge), Oscar Branson (University of Cambridge), Adam Subhas (WHOI)

Brief organization description, including organization type (e.g., for-profit, NGO, research institution/university) <20 words

Public University and Research Institution

1. Public summary of proposed project[1] to Frontier

a. **Project title** <20 words

OAE-mix: Quantifying secondary precipitation and sampling strategies in the near-field mixing zone.

[1] We use “project” throughout this template, but the term is not intended to denote a single facility or experimental setup. The “project” being proposed to Frontier could include multiple facilities/locations for research activities.

b. **Project summary and field impact:** Provide an overview of your core research goal and the benefits you think it could provide to the broader CDR field. Tell us how your project meaningfully advances knowledge on at least one of the priority innovation areas identified in the RFP. What needs to happen for your project to unlock step-change advancements in cost and scale for this innovation area? Please include figures and system schematics, and be specific but concise. <1000 words

Ocean alkalinity enhancement (OAE) is a marine carbon dioxide removal (mCDR) approach with the potential to permanently sequester atmospheric CO_2 at gigatonne scale. Characterizing the 'near-field' region within ~10km of the alkalinity release site is vital for cost-effective at-scale OAE deployments with minimal adverse environmental impact. We will combine ultra high-resolution simulations of the near-field region with data from recent field trials and new laboratory experiments to (1) quantify efficiency loss through the formation, sinking, and dissolution of secondary precipitation and (2) identify optimal deployment and sampling strategies to maximize OAE efficiency and minimize sampling costs.

Despite success in early trials, several critical bottlenecks have emerged that could limit the scaling of OAE deployments. First, secondary mineral precipitation can remove a significant fraction of the added alkalinity, reducing the efficiency of OAE. Since secondary precipitation requires high pH (with levels that depend on the mineral species involved), it occurs primarily in the near-field region before dilution occurs through mixing. Without a mitigation strategy, secondary precipitation could limit the maximum alkalinity dosage. Alkalinity release strategies could be designed to minimize this risk. This proposal will improve our understanding of secondary precipitation in the near-field mixing zone by targeting the dissolution and sinking of secondary precipitates.

A second major bottleneck for economically viable OAE is the cost of monitoring ocean deployments. Field trials conducted to date have used intensive sampling to characterize the effectiveness and environmental impacts of OAE. Field measurements are expensive, and economically viable monitoring, reporting, and verification (MRV) for OAE will require highly optimized sampling instrumentation and strategies to minimize uncertainty and costs. We will use observing system simulation experiments (OSSEs) in ultra high-resolution (~1m) near-field models to design and test optimal sampling strategies in the near-field region.

Existing field trials provide valuable insight into the near-field mixing region. The LOC-NESS project (Locking Ocean Carbon in the Northeast Shelf and Slope) completed a field trial in 2025 with the release of tracer dye and an alkaline solution in the Gulf of Maine. A wealth of data was collected from ships, sensors, and autonomous platforms to track carbon uptake and assess environmental impacts. Preliminary analysis suggests that ~15% of the added alkalinity may have been lost to secondary precipitation of magnesium hydroxide. We will leverage this unique data set to develop, test, and validate a 'digital twin' of the near-field mixing zone that can capture meter-scale variability and losses through secondary precipitation. The resulting model will provide a testbed for alkalinity release and sampling strategies that will benefit the entire mCDR industry.

A loss of dissolved alkalinity via the formation and sinking of secondary precipitation was measured in the LOC-NESS trial. Through laboratory tests and deckboard incubations, the phase appears to be mostly magnesium hydroxide ($Mg(OH)_2$). However, the fate of this particulate alkalinity is unknown. Since magnesium hydroxide readily dissolves in seawater, the precipitated particles are likely to re-dissolve in the water column. This process re-introduces dissolved alkalinity to the system, but identifying where dissolution occurs in the water column is critical to quantifying the impact on OAE efficiency. The balance of dissolution (alkalinity production) and sinking (alkalinity

removal) thus emerges as a universally critical process for not only particulate feedstocks such as brucite and slaked lime slurries, but also for sodium hydroxide-based (electrochemical) OAE.

Quantifying dissolution and sinking require microscale measurements of particle size, shape and interaction with the flow physics that are difficult to make *in situ*. Here, laboratory experiments capable of capturing the dissolution, sinking velocities, and particle sizes of secondary precipitates will provide valuable insights, will help interpret field data, and will contribute to the construction of models that can generalize these phenomena to other OAE deployment scenarios.

Our project will address these key challenges by achieving three overall objectives:

- Quantify sinking and dissolution rates of secondary precipitates formed in the near-field mixing zone.
- Find alkalinity release strategies that minimize secondary precipitation.
- Identify optimal sampling strategies to minimize uncertainty and MRV cost.
- Publish a validated 'digital-twin' of the near-field mixing zone, allowing others to test deployment and sampling strategies.

Our project addresses issues at the heart of two of Frontier Climate's Target Innovation Areas:

1. **Strategies to maximize alkalinity addition rates for Ocean Alkalinity Enhancement (OAE)**
   - New laboratory experiments will characterize the particle size, composition, sinking, and dissolution of secondary magnesium hydroxide particles.
   - An ultra-high resolution (~1m) 'digital twin' of the near-field mixing zone using new capabilities provided by the OceanBioME+Oceananigans models, including secondary mineral sinking and dissolution will be validated using field data collected during the 2025 LOC-NESS field trial.
   - Alternate scenario numerical experiments, varying e.g. alkalinity release rate, deployment time, discharge depth, will be used to identify strategies to minimize alkalinity loss through secondary precipitation.
2. **Improving the MRV toolkit for open-system pathways**
   - The validated 'digital twin' will provide a platform to virtually test sensor combinations and sampling strategies, minimizing MRV costs of future deployments, while maintaining a high level of confidence in safety and efficacy.
   - Realistic near-field models make up a key part of the MRV modelling framework, providing source conditions for regional and global scale models.
   - Generation of particle size/sinking/dissolution relationships will unlock *in situ* sampling strategies for quantifying net alkalinity addition using currently available sensor technology.

For OAE to scale credibly, operators must be able to demonstrate net alkalinity addition (accounting for losses to secondary precipitation) using cost-effective, field-deployable monitoring systems. This requires: (i) a quantitative understanding of particle fate in real deployment conditions; (ii) models capable of representing these processes at the spatial and temporal scales relevant to near-field mixing; and (iii) optimized monitoring strategies that minimize cost while maintaining confidence in CDR accounting. This project delivers all three.

a. **Project objectives:** Describe your project in more detail: what are your key research questions and proposed approach for answering them, including methods, project location, and scale where

applicable? Tell us why your approach is best-in-class, and how you're differentiated from any other organization working on a similar approach. Please include figures and system schematics, and be specific but concise. 1000-1500 words

This project advances Frontier's priority innovation areas for mCDR, specifically: assessing the influence of secondary precipitation on OAE efficiency, and developing robust MRV strategies to guide the mCDR industry. By delivering open, validated model output and proposing practical monitoring protocols, the project lowers barriers to deployment and credible accounting across the mCDR industry.

Research questions

Our project addresses six research questions:

- What are the size distributions, settling velocities, and dissolution rates of secondary precipitates formed during point-source OAE deployments?
- How do particle dynamics (sinking, resuspension, dissolution) affect the net alkalinity budget in the near-field mixing zone?
- What is the net impact of secondary precipitation on OAE efficiency across a range of ocean conditions and deployment strategies?
- How does upper ocean turbulence interact with particle settling through resuspension and entrainment to determine the fate of precipitated material?
- Which alkalinity release strategies minimize alkalinity loss through secondary precipitation under a wide range of environmental conditions?
- What sampling strategies and sensor combinations minimize uncertainty in the near-field region while reducing monitoring costs?

Our approach

We will answer these questions and meet the objectives set out above using three complementary approaches:

- Analysis of the 2025 LOC-NESS field trial dataset, including *in-situ* and shipboard measurements of particle size distributions, carbonate chemistry, and alkalinity evolution following point-source addition in the Gulf of Maine. This unique dataset will be used to calibrate and validate the near-field model.
- Laboratory experiments will directly measure particle size distributions, settling velocities, and dissolution rates for secondary precipitates under controlled conditions relevant to OAE deployments. A recently developed sensor system provides directly comparable particle measurements in both *ex-situ* (laboratory) and *in-situ* (field) settings.
- High-resolution near-field modelling with Oceananigans and OceanBioME will simulate the near-field mixing zone at ~1m resolution with realistic biogeochemistry and particle dynamics. OceanBioME's 'active particle' capability enables computationally efficient simulation of particle sinking, resuspension, and dissolution.

These approaches are described in more detail below.

Analysis of the 2025 LOC-NESS field trial dataset

The LOC-NESS project, led by WHOI, conducted the first EPA-permitted open-water field trial of ocean alkalinity enhancement (OAE) in US waters. The trial used a liquid alkalinity feedstock, sodium hydroxide (NaOH), dispersed into the Gulf of Maine. A critical and incompletely understood consequence of introducing liquid NaOH into seawater was the formation of secondary precipitation: solid magnesium hydroxide particles that nucleated and grew as the highly alkaline effluent mixed with ambient seawater. This precipitation is distinct from the intended dissolved-alkalinity signal and introduces a particulate phase with its own dynamics.

The fate of these secondary precipitates is central to the efficiency and MRV of liquid OAE approaches. Two processes compete after particles form:

- Settling: particles sink out of the mixed layer before dissolving, transporting alkalinity to depth where it may become isolated from the atmosphere and unable to drive near-term CO_2 uptake.
- Dissolution: particles dissolve back into solution, contributing their alkalinity to the water column and enabling CO_2 drawdown if the water is re-exposed to the sea surface.

The LOC-NESS field trial generated valuable observational data on the dissolved alkalinity signal, but important details of the particles, including the particle size distribution, settling velocity spectrum, and dissolution timescale, were not directly characterized *in situ* during the trial. This is a critical gap: without knowing the properties of the secondary precipitates, the efficiency of liquid OAE cannot be rigorously bounded, and models cannot be properly constrained. Furthermore, identification of this process even for liquid-based OAE highlights the central importance of particle sinking and dissolution for nearly all OAE approaches. The proposed research directly addresses this gap.

Laboratory characterization of particle setting and dissolution

The Real-Time Size and Settling Velocity (RTSSV) sensor is, the only commercial instrument capable of simultaneously measuring particle size distributions and settling velocities both in situ and ex-situ processing (Figure 1). We will utilize a RTSSV in controlled laboratory experiments to directly characterize the secondary precipitation generated by liquid OAE feedstocks under conditions representative of those encountered in the LOC-NESS field trial.

The experimental program will systematically vary parameters that influence secondary precipitation properties:

- Dilution ratio: Mixing different amounts of sodium hydroxide into seawater will mimic the range of mixing conditions from near-field (high concentration) to far-field (well-mixed) in the LOC-NESS plume.
- Temperature: Conducting experiments at one of the mixing ratios and three characteristic temperatures (lower than ambient, ambient, and elevated) will evaluate the effect of temperature on particle size, dissolution, and settling velocities.
- Mixing intensity: low, medium, and strong turbulence levels at fixed temperature and mixing ratio will assess how mixing affects the fate of secondary precipitates.

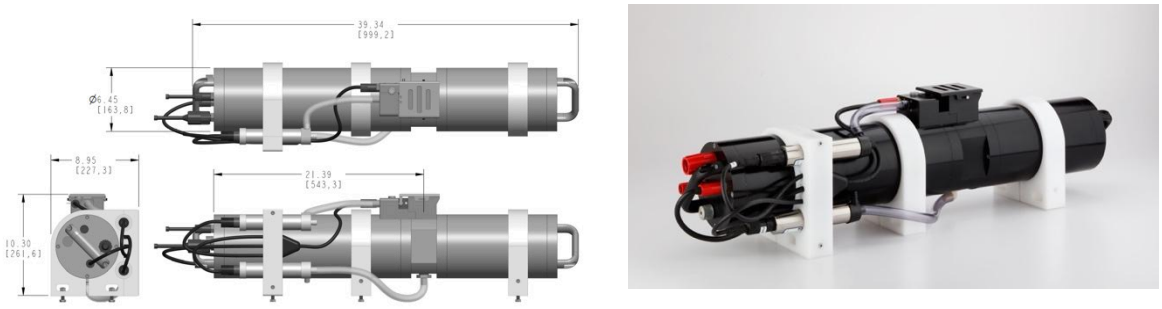


Figure 1: The LISST-RTSSV, the only instrument capable of simultaneously measuring particle size distributions and settling velocities ex-situ and in-situ.

For each experimental condition, the RTSSV will provide direct measurements of the particle size distribution and the joint size–settling velocity distribution, capturing expected substantial deviations from Stokes settling due to aggregation or irregular particle morphology. Dissolution rates will be inferred by tracking particle disappearance over time under known chemistry conditions and also by measuring bulk pH signals in the experimental chamber. Collectively, these measurements will produce a parameter library of secondary precipitation properties as a function of near-field mixing conditions, a dataset that does not currently exist for any liquid OAE system. This parameter library unlocks the potential for developing *in situ* observational programs to better characterize these processed during OAE field trials.

High resolution near-field modelling

Ocean models are a vital component of MRV for mCDR. High resolution near-field models, capable of capturing meter-scale variability are needed to assimilate measurements made near the intervention site, quantify alkalinity loss through secondary precipitation, and help assess possible environmental impacts. The output from the near-field models can then be averaged to provide source conditions at the appropriate resolution for regional and global models to assess the spatial patterns and timing of CO_2 uptake. Our project will develop near-field models that are supported and validated with field measurements, building confidence in safety and efficacy assessments of OAE.

The gold-standard method for near-field modelling are large-eddy simulations (LES). By definition, LES explicitly resolves the largest 3D turbulent motions. In the context of mCDR this has two key advantages: LES captures the dominant scales of variability in concentration fields (alkalinity, pH, etc.), and LES captures the physical processes that transport material throughout the water column. In contrast, regional models and dilution models produce overly smooth fields in the near-field environment.

The Ocean Biogeochemical Modelling Environment (OceanBioME) is a new open-source biogeochemical model led by the University of Cambridge. It allows rapid prototyping of model components and implementation in near-field, regional, and global settings using the Oceananigans ocean model developed at MIT. OceanBioME and Oceananigans are designed to run quickly on CPU computing systems, enabling inexpensive LES of the near-field region.

OceanBioME does not yet have the capability to model the fate of particles formed during deployment. We will use data collected from the lab experiments to include precipitates and we will validate the model with LOC-NESS observations. OceanBioME has a unique feature called ‘active particles’ which will allow us to simulate dissolving and settling particles in a Lagrangian framework. Importantly, active particles in

OceanBioME can interact with the tracer-based biogeochemistry, allowing dissolving particles to release alkalinity back into the water column as they sink, while the sinking speed will evolve in time, along with particle size.

Guided by the laboratory experiments and field trial data, we will build a realistic, high-resolution (~1m) near-field model of the 2025 LOC-NESS OAE deployment in the Gulf of Maine. This model will allow us to quantify the fate of secondary precipitates and the associated OAE efficiency loss and explore optimal alkalinity release and sampling strategies. The model will be made publicly available for the OAE community to use to test additional sensor platforms and sampling strategies.

b. **Risks:** What are the biggest risks to your project's success and how will you mitigate those? Include scientific; technical; project execution; measurement, reporting and verification (MRV); ecosystem; financial; and any other risks. <500 words

The primary scientific risk lies in representing secondary particle dissolution and sinking in a computationally efficient modelling framework. Capturing fast reaction kinetics and particle processes may require small timesteps, limiting simulation scalability.

We mitigate this risk through a multi-tiered modelling strategy:

- Develop a high-fidelity model calibrated against laboratory and field data
- Derive reduced-complexity parameterizations from this model
- Validate reduced-complexity models against observations to ensure accuracy

Our approach ensures delivery of scientific insight and a practical, deployable modelling tool, even if full-complexity simulations prove computationally expensive.

Secondary precipitation may involve multiple mineral phases forming simultaneously with different properties. While we will focus on $Mg(OH)_2$ precipitates formed from the mixing of NaOH and seawater, our approach combining lab experiments, field data, and near-field modelling will provide a framework to characterize other mineral species. Our characterization of the sinking and dissolution of brucite particles will also provide insight into solid-phase $Mg(OH)_2$ OAE feedstock including the $Mg(OH)_2$ slurry used by Planetary Technologies.

Collaboration across multiple institutions and engagement with the mCDR industry introduces coordination risk. We will fund a project coordinator (10% time), leveraging personnel and expertise at the Centre for Climate Repair which is based at the University of Cambridge. The project coordinator will coordinate data sharing and communication both internally with the project team and with external stakeholders and beneficiaries. Close collaboration of the project team will be ensured through joint meetings from project outset, including an in-person kick-off meeting at the University of Cambridge. A workshop will be held at the end of the project meeting to engage with the mCDR industry and to translate the science into practice. Each workstream (laboratory, modelling, data analysis) has a designated lead with explicit deliverables, and our multi-pronged strategy ensures standalone value for each component.

As an academic consortium, the project is not exposed to commercial financial risk. The low overhead limit might lead to under-recovery and require financial support from our institution to cover local costs.

c. **Proposal to Frontier:** Please list your primary research deliverables, project timeline, and requested funding below. If you are selected for Frontier funding, this table will form the basis of contract discussions.

Primary research deliverables *(List tangible outputs of the project: e.g., validated datasets, peer-reviewed models, bench-scale prototypes, or field trial results. Should focus on resolving a specific bottleneck or uncertainty identified in Section 1.a)*	• Laboratory dataset: secondary precipitate properties (size, composition, settling velocity, dissolution rate) across OAE-relevant conditions • OceanBioME with validated solid-phase alkalinity (sinking, resuspension, dissolution) • Open, publicly released near-field model of the 2025 LOC-NESS field trial with full alkalinity budget, including secondary precipitation losses • Evidence-based guidance on cost-effective sensor configurations, sampling strategies
Field impact *(Briefly describe how these deliverables resolve key bottlenecks for a specific CDR pathway or broader CDR field, e.g. efficiency gains, scaling unlocks, or reducing measurement uncertainty.)*	This research directly reduces two critical sources of uncertainty in OAE CDR accounting: (1) alkalinity loss through secondary precipitation, and (2) MRV cost and uncertainty in the near-field mixing zone. Validated parameterizations and open modelling tools will lower barriers to scientific and commercial deployment, enabling more accurate CDR accounting across the OAE industry. Optimal monitoring protocols identified from observing system simulation experiments will reduce operator MRV costs while maintaining confidence in safety and efficacy.
Project timeline & key milestones *(At what point should Frontier consider the research project complete? Include major progress markers. Frontier generally prefers projects with clear readouts within ~18 months.)*	**Project duration:** October 1, 2026 –September 30, 2028 (24 months) **Q2 (April 2027):** Implement framework to model solid-phase alkalinity in OceanBioME

	Q6 (April 2028): Final laboratory characterization of secondary precipitation; LOC-NESS near-field simulation complete and validated **Q8 (September 2028):** OSSE analysis and MRV optimization guidance complete; near-field simulation data published
Total grant request (US$) *(The total funding required to execute the research scope described above. Frontier R&D grants typically range from US$250k–$750k.)*	**$750,000** over 24 months (1 Oct 2026 – 30 Sept 2028)